

| Q1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Q2                                                                                                                                                                                                                                                                                                                                                         | Viva Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Build a Course Enrollment application in React where students can enroll in additional/summer term courses. Use input fields to collect the student name and course name, and display an "Enroll Now" button. Use the useState hook with an object state variable (useState({})) to manage form data and enrollment count dynamically. Display enrolled student details and total enrollment count. Implement validation to prevent empty submissions. Use reusable event handling logic using the name attribute. | Testing: Create a Login page in React with email and password fields. When the user enters valid credentials, display a success message; otherwise, display an error message. Write unit test cases using React Testing Library and Jest to test both successful and invalid login scenarios.                                                              | <ol style="list-style-type: none"> <li>1. Why is a controlled component preferred in forms engineering?</li> <li>2. What is the stale state problem? How can it be avoided?</li> <li>3. What is the purpose of an Error Boundary? What happens if there is no boundary?</li> <li>4. Differentiate between Imperative programming and Declarative programming.</li> <li>5. What is the benefit of having cache memory with API calls?</li> </ol>                     |
| Build a Recipe Blog UI application in React using Tailwind CSS and fetch recipe data from the DummyJSON API. Display recipes in a responsive card layout showing recipe image, title, cook time, difficulty level, rating, and a "View Recipe" button. Use useState and useEffect for data fetching. Display Skeleton UI cards while fetching. Implement cache memory logic to prevent multiple unnecessary API calls.                                                                                             | Lifted State: A currency converter application has two input boxes — USD and INR. When the user enters an amount in one box, the converted value should appear in the other. State should be managed in the parent component and passed to child components through props. Write a React application to implement this using lifted state.                 | <ol style="list-style-type: none"> <li>1. What is the purpose of the useState hook? Why do we need it?</li> <li>2. When should state be lifted to the parent component? Explain derived state.</li> <li>3. Explain the benefits of using Tailwind CSS instead of traditional CSS.</li> <li>4. Explain the purpose of protected routing.</li> <li>5. What is side-effect code in React? When do we use the useEffect hook? Write the syntax of useEffect.</li> </ol> |
| Create an Expense Tracker application in React using useState[] and useEffect. Create input fields expense_title and amount inside an object state variable (useState({})). Add buttons to Add and Delete Expenses. Use Tailwind CSS. Use localStorage to store data and useEffect to load on start. Display expense title, amount, and total dynamically. Implement validation for empty/invalid entries.                                                                                                         | API Cache Race Condition: A shopping website has a product search box. When the user types, matching products should be fetched from an API. If the same product is searched again, load from cache using useRef(). Cancel previous API requests when the user types fast to avoid showing incorrect results. Write a React application to implement this. | <ol style="list-style-type: none"> <li>1. Why do we perform testing? Differentiate unit testing and integration testing.</li> <li>2. Are props mutable or immutable? Is props flow unidirectional or bidirectional? Justify your answer.</li> <li>3. Why do we deploy or host a developed project in the production phase?</li> <li>4. Write the React rendering UI pipeline.</li> <li>5. Differentiate between Container and Presentational components.</li> </ol> |

| Q1                                                                                                                                                                                                                                                                                                                                                                                                                                    | Q2                                                                                                                                                                                                                                                                                                                                                                                           | Viva Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Create a Product Card application in React using functional components and useState. Store product name, price, discount percentage, and quantity inside an object state variable in App and pass as props to ProductCard. Create buttons to Apply Discount and Toggle Wishlist. Display discounted price dynamically. Use conditional rendering to show "Out of Stock" if quantity is 0, otherwise show "Add to Cart" button.</p> | <p>Context API: A college student portal has Navbar, Dashboard, and Attendance Panel components. Student name should display in Navbar and attendance percentage in Attendance Panel. Create a separate Context file to share data globally. Write a React application using Context API so any component can access student data directly without prop drilling.</p>                        | <ol style="list-style-type: none"> <li>1. What is the purpose of TypeScript? Differentiate between Interface and Type Alias.</li> <li>2. Which library is installed to support routing? Explain SPA and why routing is required.</li> <li>3. List the common styles generally applied on a button.</li> <li>4. Explain race condition in the context of a search API endpoint.</li> <li>5. Explain Conditional Rendering in React with an example. If the state variable errorMsg is not empty, display the error message using the tag.</li> </ol> |
| <p>Create a Food Menu application in React using functional components and props. Display food image, food name, price, category, rating, and availability status inside styled cards. Maintain all details as a list of objects. Create a reusable FoodItemCard component and reuse it to display different food items received through props.</p>                                                                                   | <p>Testing: Create a React Password Strength Checker component where the user enters a password and the application validates minimum length, uppercase letter, lowercase letter, digit, and special symbol. Display appropriate validation messages or a strong password message. Write unit test cases using React Testing Library and Jest to test all password validation scenarios.</p> | <ol style="list-style-type: none"> <li>1. Explain DOM vs Virtual DOM and what is reconciliation.</li> <li>2. What is Virtual DOM (VDOM)? When will it be created and re-created?</li> <li>3. What is prop drilling? Explain the steps to create global state using Context API.</li> <li>4. Write examples to differentiate between pure and impure JavaScript functions.</li> <li>5. Why dynamic routing is required? How is the path specified and which hook retrieves parameters?</li> </ol>                                                    |

| Q1                                                                                                                                                                                                                                                                                                                                                                                                                                    | Q2                                                                                                                                                                                                                                                                                                                                                                                                             | Viva Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Build a Product Store UI application in React using Tailwind CSS and fetch product data from DummyJSON API. Display products in a responsive card layout showing product image, title, price, category, rating, stock availability, and "Add to Cart" button. Use useState and useEffect for data fetching. Display Skeleton UI product cards while fetching. Implement cache memory logic to prevent multiple unnecessary API calls. | Lifted State: A WhatsApp-like chat application has two components — one for typing a message and another for displaying the live preview. State is managed in the parent and shared via props. Display: total characters entered and remaining characters allowed out of 100. Write a React application using lifted state and derived state.                                                                  | <ol style="list-style-type: none"> <li>1. What is the purpose of Skeleton UI while fetching data from an API?</li> <li>2. What is the benefit of having a separate service file for API fetching?</li> <li>3. Explain the purpose of the useRef hook. Write the steps to get DOM reference and read the text entered by the user in an input element.</li> <li>4. What is Skeleton UI and what is its purpose?</li> <li>5. Assume an object course with fields courseName, duration, and instructor is passed as props. How to access the received props? Can the child component modify the received props?</li> </ol> |
| Create a Course Dashboard application in React using Tailwind CSS. Display separate components for CourseDetails, VideoLectures, and ProgressTracker. Implement an ErrorBoundary component to catch rendering errors inside the VideoLectures component. Display a reusable fallback UI component when the video API service fails.                                                                                                   | Testing: Create a React shopping application with ProductList and Cart components. When the user clicks "Add to Cart" in ProductList, the product should appear in Cart and the cart count and total price should update. Write integration test cases using React Testing Library and Jest to test interaction between ProductList and Cart for adding products, updating totals, and handling an empty cart. | <ol style="list-style-type: none"> <li>1. How does the Container-Presenter pattern help React developers?</li> <li>2. What is Reconciliation in React?</li> <li>3. What happens if no Error Boundary exists in a React application? How is the Error Boundary component helpful?</li> <li>4. Differentiate between Full Page Load and Single Page Application (SPA).</li> <li>5. Write the commands to: Create a React application and Run the local development server.</li> </ol>                                                                                                                                     |

| Q1                                                                                                                                                                                                                                                                                                                                                                                                                                       | Q2                                                                                                                                                                                                                                                                                                                                                                                                                                 | Viva Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Create a Payroll Calculator application in React using functional components and useState. Create input fields for Employee Name, Basic Pay, DA%, and HRA%. Assume PF = 10% of Basic Pay and Tax = 5% of Gross Salary. Take state variable as object. Use onClick buttons to calculate Gross Salary, Tax Amount, Total Deductions, and Net Salary. Display all calculated values dynamically. Do validation for all input fields.</p> | <p>Testing: Create a React search application with SearchBar and ResultsList components. When the user types in SearchBar, ResultsList should display only matching items. If no matching items found, display "No results found". Write integration test cases using React Testing Library and Jest to test interaction between SearchBar and ResultsList.</p>                                                                    | <ol style="list-style-type: none"> <li>1. When do you prefer CSS Modules over Global CSS? What are the benefits of CSS Modules? Write the steps to create and use a CSS Module.</li> <li>2. Write the syntax for creating a function-based React component with default export and relevant import statement.</li> <li>3. How do you decide between Controlled Components and Uncontrolled Components?</li> <li>4. If you style a card component (Profile/Course/FoodItem/Product Card), what styles are generally applied to make the UI look like a card?</li> <li>5. What is the reason behind maintaining closure memory in JavaScript?</li> </ol>                                                                                                                       |
| <p>Create a Theme Toggle application in React using functional components and useState. Create a button to switch between Light Theme and Dark Theme dynamically. Use localStorage to save the selected theme and useEffect to load the stored theme on start. Apply conditional rendering and dynamic class names to change the application theme UI.</p>                                                                               | <p>Routing: Create a student portal website using React Router DOM with Home, Login, Courses, and Dashboard pages. When a student selects a course, open the Course Details page based on course ID in the URL. Inside Dashboard, Marks and Attendance pages should render within the Dashboard. Only logged-in users can access Dashboard pages; otherwise redirect to Login. Display a Page Not Found page for invalid URLs.</p> | <ol style="list-style-type: none"> <li>1. Explain optional fields in a TypeScript object structure. What pre-check should be done before accessing an optional field?</li> <li>2. How does the Component Composition pattern help React developers? Justify the reason behind creating multiple components instead of a single component.</li> <li>3. Write productService.js using Promise Chaining with proper error handling to fetch data from: <a href="https://dummyjson.com/products">https://dummyjson.com/products</a></li> <li>4. Explain Union Types and Union Values in TypeScript with example. What pre-check should be done before applying type-specific operations?</li> <li>5. Explain the purpose of the useRef hook in a real-world scenario.</li> </ol> |

| Q1                                                                                                                                                                                                                                                                                                                                                                                        | Q2                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Viva Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Create a Todo List application in React using useState[]. Create an input field and buttons to add and remove tasks. Use Tailwind CSS for styling. Use localStorage to store tasks and useEffect to load stored tasks on start. Display an empty message when no tasks are available and prevent empty task submission.</p>                                                            | <p>Testing: Create a React Product Card component with a Wishlist button. When clicked, the product should be added to or removed from the wishlist, the heart icon should change, and a message should display. Write unit test cases using React Testing Library and Jest to test the initial button state, button toggle behavior, and wishlist messages.</p>                                                                                                  | <ol style="list-style-type: none"> <li>1. What is the state state problem in React? How can it be avoided? Explain with a useEffect example.</li> <li>2. Explain race condition in the context of a search API endpoint. How can it be resolved using cleanup functions?</li> <li>3. Write the syntax for creating a function-based React component with default export. What is the relevant import statement?</li> <li>4. Explain optional fields in a TypeScript object structure. What pre-check should be done before accessing an optional field?</li> <li>5. How does the Container-Presenter pattern help React developers? Justify the reason behind creating multiple components.</li> </ol>                |
| <p>Create a Book Card application in React using functional components and useState. Store book title, price, discount percentage, and stock quantity inside an object state variable in App and pass as props to BookCard. Create buttons to Apply Discount and Toggle Favorite. Use conditional rendering to show "Out of Stock" if quantity is 0, otherwise show "Buy Now" button.</p> | <p>Routing: Create a blog website using React Router DOM with Home, About, Blog, Login, and Dashboard pages. When a user selects a blog post, open the Blog Details page based on blog ID in URL. Inside each blog page, Comments and Related Posts should render within the same page. Dashboard should contain Profile and Settings pages. Only logged-in users can access Dashboard; otherwise redirect to Login. Display Page Not Found for invalid URLs.</p> | <ol style="list-style-type: none"> <li>1. What is the purpose of an Error Boundary? What happens if no Error Boundary exists in a React application? How can it be implemented?</li> <li>2. Differentiate between Full Page Load and Single Page Application (SPA). Why is routing required and which library supports it?</li> <li>3. What is prop drilling? Explain how Context API solves the prop drilling problem with steps to implement it.</li> <li>4. Explain Union Types in TypeScript with an example. What pre-check should be done before applying type-specific operations?</li> <li>5. Why do we perform testing? Differentiate between unit testing and integration testing with examples.</li> </ol> |